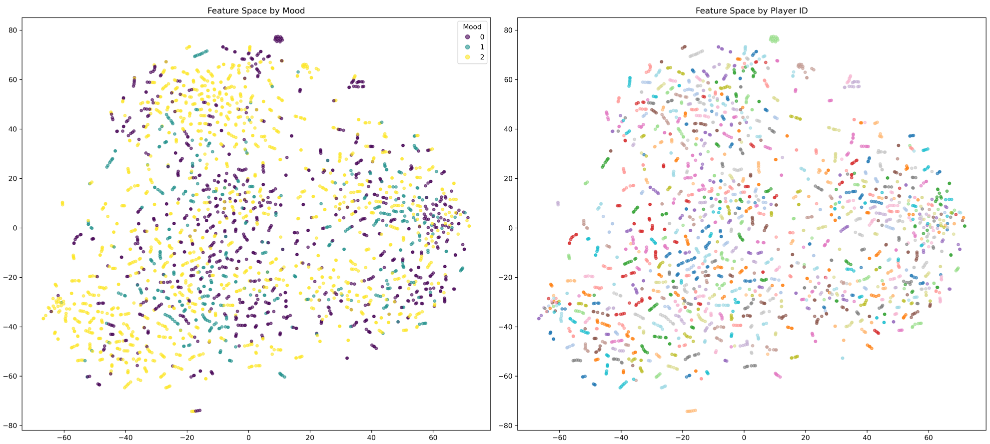


Figure 1: t-SNE visualization for Solid (Fold 1). Points are colored by Player ID, showing high intermixing across players. Similar patterns were observed in all 9 games and across all 5 folds.

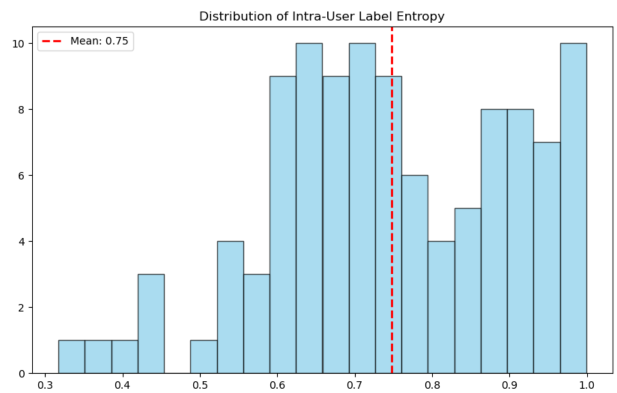


Figure 2: Distribution of intra-user label entropy. The mean of 0.75 indicates substantial label diversity within individual players' sessions.

1 data split evidence examples

Here we provide supplementary evidence supporting the data split protocol described in Section 4.2 of the main paper. All analyses are based on the Solid dataset as an example shown here. The evidence of other game datasets can be found in our supplementary repository.

1.1 t-SNE Visualization of Feature Embeddings

Figure 1 shows t-SNE visualizations of feature embeddings extracted from the Solid test set using the full fine-tuned ViViT model, colored by Player ID. The consistent intermixing across players in all games supports our claim that player identity does not create distinct clusters in the feature space.

1.2 Distribution of Intra-User Label Entropy

Figure 2 presents the distribution of intra-user label entropy across all players in Solid. The mean entropy of 0.75 confirms high label diversity within individual players' sessions.

Table 1: Cross-val results of game performance by layer (weighted F_1). All values are percentages. Best results per game are in **bold**.

Layer	apex	endless	fps	gallery	gun	platform	solid	tiny	topdown
embeddings	50.70	46.23	58.15	45.92	49.10	41.29	58.58	48.39	53.40
layer.0	**72.97**	**61.49**	**72.67**	**65.41**	**68.24**	**72.88**	**73.19**	**64.97**	**71.46**
layer.1	65.79	55.84	71.81	62.07	56.28	60.18	72.86	55.79	68.42
layer.2	56.06	53.30	68.08	61.97	51.90	62.58	70.01	54.41	62.59
layer.3	56.09	52.36	69.70	58.68	50.09	54.73	65.30	51.49	61.17
layer.4	56.28	50.67	68.18	58.17	48.49	52.16	62.25	50.49	61.35
layer.5	54.02	53.48	67.66	59.50	50.03	52.88	60.86	54.21	61.87
layer.6	48.30	53.68	70.16	55.80	48.79	54.68	61.38	54.03	61.47
layer.7	49.30	54.11	70.20	60.79	50.08	52.02	62.13	53.78	63.69
layer.8	47.76	51.70	70.23	56.55	52.07	55.56	62.40	52.80	62.75
layer.9	52.27	53.84	72.36	61.06	52.90	54.57	64.08	53.41	63.54
layer.10	50.00	54.72	70.56	58.52	51.76	58.73	67.22	54.25	64.96
layer.11	47.96	50.57	67.15	52.89	48.31	49.25	57.92	52.86	60.46
classifier only	40.27	41.12	55.18	40.79	37.94	42.13	46.71	45.43	50.26

Table 2: Cross-validation results of game performance by layer (macro F_1). All values are percentages. Best results per game are in **bold**.

Layer	apex	endless	fps	gallery	gun	platform	solid	tiny	topdown
embeddings	44.0	40.76	52.23	32.8	42.61	32.13	50.68	38.43	44.67
layer.0	**71.18**	**58.8**	**69.91**	**60.75**	**65.29**	**71.24**	**69.54**	**62.71**	**67.61**
layer.1	63.83	52.01	68.22	55.72	50.61	54.87	67.29	48.38	65.29
layer.2	50.7	49.72	63.57	55.85	44.74	58.62	66.62	46.5	57.36
layer.3	50.84	48.07	65.93	51.41	42.95	50.07	59.36	44.36	54.98
layer.4	51.55	45.96	63.96	50.15	40.23	45.03	55.31	43.04	54.86
layer.5	48.92	49.35	63.34	52.65	42.5	47.81	54.27	46.67	56.16
layer.6	42.0	50.01	66.62	47.16	40.26	50.07	54.9	46.67	55.55
layer.7	41.95	50.7	66.42	54.21	42.68	45.58	56.92	46.92	58.54
layer.8	40.97	47.56	66.3	48.25	45.12	49.95	56.91	44.63	57.67
layer.9	46.63	50.52	68.99	54.34	45.87	49.42	59.62	45.57	58.91
layer.10	43.99	51.08	66.8	50.99	44.86	53.98	62.32	46.81	60.43
layer.11	40.22	46.33	62.96	43.42	40.65	42.72	51.25	45.34	54.91
classifier only	30.93	34.06	45.63	24.07	27.24	32.87	34.53	35.46	39.65

2 Empirical Validation of the Selection Criterion

To assess the reliability of our proposed L2 selection criterion, we conducted a preliminary study comparing it against the exhaustive cross-validation (CV) oracle used in Surgical Fine-Tuning (SFT) [1]. This SFT-CV protocol, which involves individually fine-tuning each candidate layer (or block) to select the best-performing one based on validation performance, was shown in the original work to be the most effective selection strategy, outperforming their proposed heuristic criteria (RGN/SNR).

Protocol. We implemented this SFT-CV protocol on a subset of games under a simpler data split for computational feasibility. Each candidate layer (alongside the classifier) was fine-tuned for a short budget of 3 epochs—a duration we found sufficient to observe significant performance differentiation. The layer achieving the highest weighted F_1-score on a held-out validation set was selected as the CV oracle choice.

Result. As shown in Table 1 and Table 2, the single layer selected by our L2 criterion during SALFT's exploratory phase **exactly matched the oracle layer identified by the exhaustive SFT-CV protocol** in all tested cases. This perfect alignment provides strong empirical evidence that our efficient, single-run L2 metric is a reliable proxy for the computationally expensive optimal selection procedure.

Table 3: Paired t-test results comparing SALFT with fully fine-tuned ViViT and the best-performing baseline, arranged by game. All values are rounded to four decimal places.

Game	Metric	Ours (Mean±Std)	vs Full (p)	Sig vs Full	Diff vs Full	vs Base (p)	Sig vs Base	Diff vs Base
apex	f1_weighted	0.5845 ± 0.0070	0.5509	ns	0.0040	0.5509	ns	0.0040
apex	f1_macro	0.5400 ± 0.0122	0.6330	ns	-0.0044	0.6330	ns	-0.0044
endless	f1_weighted	0.5558 ± 0.0137	0.1908	ns	0.0104	0.1219	ns	-0.0269
endless	f1_macro	0.5149 ± 0.0179	0.3135	ns	0.0091	0.1025	ns	-0.0340
fps	f1_weighted	0.6275 ± 0.0139	0.6691	ns	0.0038	0.5665	ns	0.0038
fps	f1_macro	0.5799 ± 0.0158	0.5985	ns	0.0053	0.7126	ns	0.0026
gallery	f1_weighted	0.5669 ± 0.0162	0.1192	ns	0.0141	0.1192	ns	0.0141
gallery	f1_macro	0.4958 ± 0.0197	0.0378	*	0.0234	0.0378	*	0.0234
solid	f1_weighted	0.6058 ± 0.0147	0.2035	ns	0.0115	0.2035	ns	0.0115
solid	f1_macro	0.5453 ± 0.0146	0.6763	ns	0.0023	0.6763	ns	0.0023
gun	f1_weighted	0.5727 ± 0.0130	0.0097	**	0.0312	0.0097	**	0.0312
gun	f1_macro	0.5408 ± 0.0131	0.0074	**	0.0324	0.0074	**	0.0324
platform	f1_weighted	0.5639 ± 0.0243	0.0823	ns	0.0202	0.0823	ns	0.0202
platform	f1_macro	0.5308 ± 0.0253	0.1222	ns	0.0218	0.1222	ns	0.0218
tiny	f1_weighted	0.5560 ± 0.0187	0.5710	ns	-0.0075	0.5710	ns	-0.0075
tiny	f1_macro	0.5211 ± 0.0256	0.4776	ns	-0.0128	0.4776	ns	-0.0128
topdown	f1_weighted	0.5974 ± 0.0120	0.5704	ns	0.0083	0.5796	ns	-0.0036
topdown	f1_macro	0.5495 ± 0.0135	0.7756	ns	0.0042	0.1485	ns	-0.0113

3 Statistical Significance Tests of SALFT

To rigorously evaluate our method, we conduct paired t-tests ($\alpha = 0.05$) for each game and each F_1 metric, comparing SALFT to: (1) fully fine-tuned ViViT and (2) the best-performing non-SALFT baseline per game and metric. Mean differences (Diff), p-values, and significance levels ($*$: $p < 0.05$, $**$: $p < 0.01$) are reported. Positive Diff indicates that SALFT outperforms the compared method.

Notation: '*' denotes $p < 0.05$, '**' denotes $p < 0.01$, and 'ns' indicates not significant. Positive Diff indicates an improvement of SALFT over the compared method.

4 Complete Computational Efficiency Analysis

In this section, Figure 3 shows the complete set of computational efficiency curves for rest eight games, complementing the representative example shown in Figure 2 of the main paper. These curves consistently demonstrate that SALFT achieves performance comparable to full fine-tuning with significantly lower cumulative computational cost across all games.

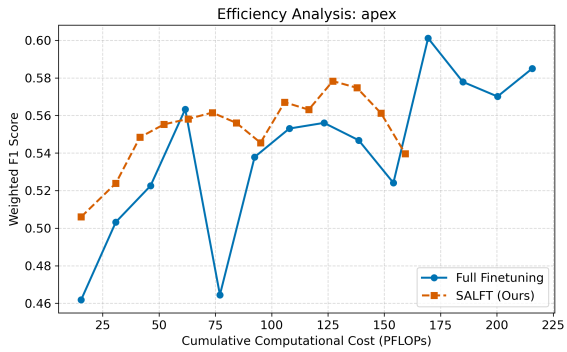
5 Comparison of Attention Patterns: SALFT vs. Full Fine-Tuning

To further understand the behavioral differences between our proposed SALFT method and standard full fine-tuning, we compare their attention patterns using representative samples from the `solid` game. Both models analyzed in this section are from the first fold of the 5-fold cross-validation.

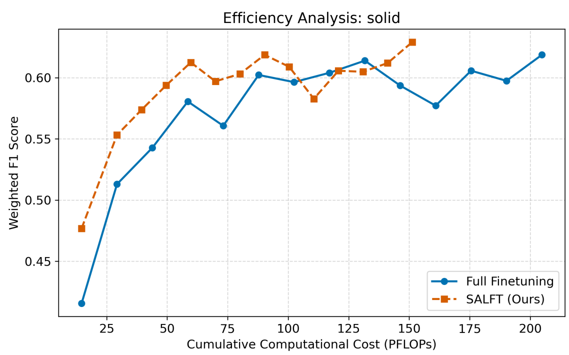
As shown in Figure 4, the model fine-tuned via full fine-tuning exhibits a strong tendency to focus **narrowly** on specific, well-defined regions (e.g., a leading vehicle or lane boundaries), producing sharply localized saliency maps. While this indicates precise attention to salient objects, it may limit the model's capacity to integrate broader contextual cues from the scene.

In contrast, the SALFT-fine-tuned model (see the Figure 3 of main paper) demonstrates a slightly broader attentional spread. It covers not only primary objects of interest but also incorporates relevant surrounding context, such as road edges, background structures, and indicators of camera motion. This pattern appears better suited for interpreting the complex, dynamic scenes characteristic of gameplay videos.

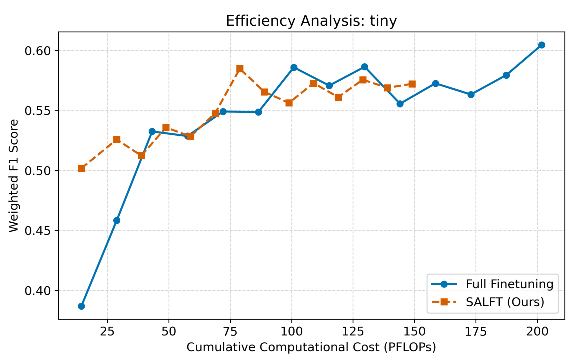
This divergence in attention strategy may underlie the performance differences noted in the main paper. The broader, context-integrating attention of the SALFT model likely contributes to its robust performance across most games. Furthermore, we observed that the full fine-tuned model occasionally showed reduced sensitivity to subtle emotional transitions, misclassifying some arousal-decreasing segments as arousal-unchanging, whereas the SALFT model maintained better discriminability in these cases. Additional atten-



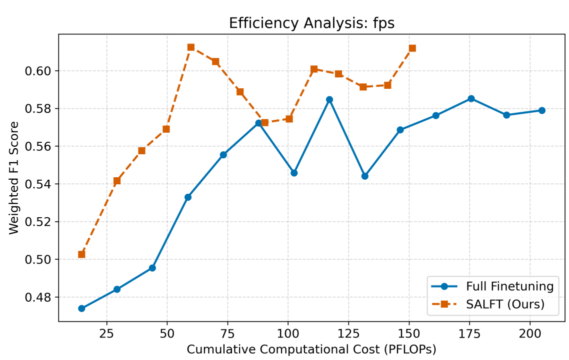
(a) `apex`



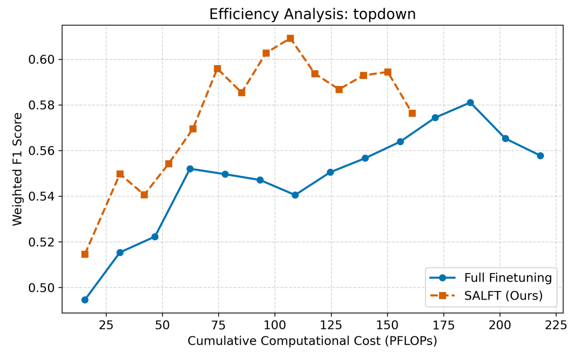
(b) `solid`



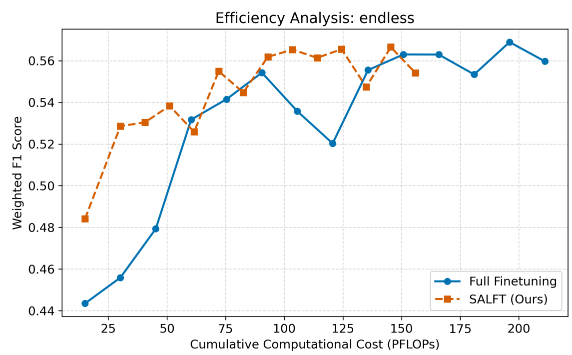
(c) `tiny`



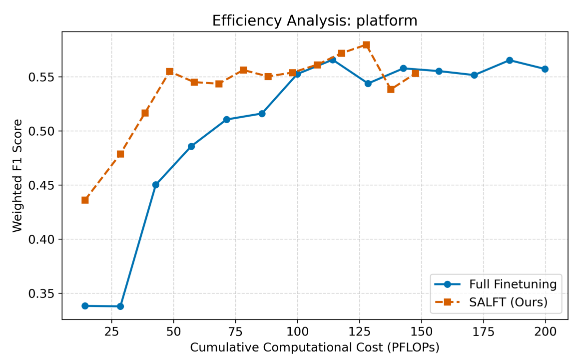
(d) `fps`



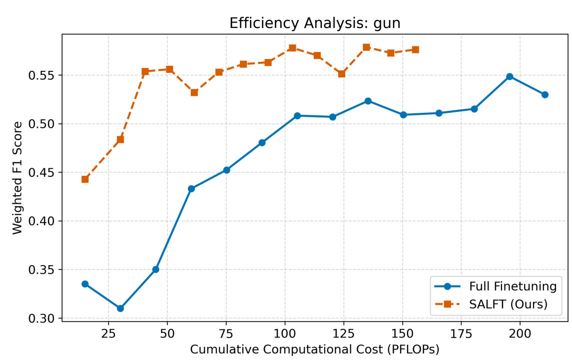
(e) `topdown`



(f) `endless`



(g) `platform`



(h) `gun`

Figure 3: Complete set of weighted F_1 against cumulative fine-tuning FLOPs for all nine games (full fine-tuning vs. SALFT). The two-stage nature of SALFT is evident in all curves, with an initial exploratory phase (first 2 epochs) followed by the efficient selective fine-tuning phase.

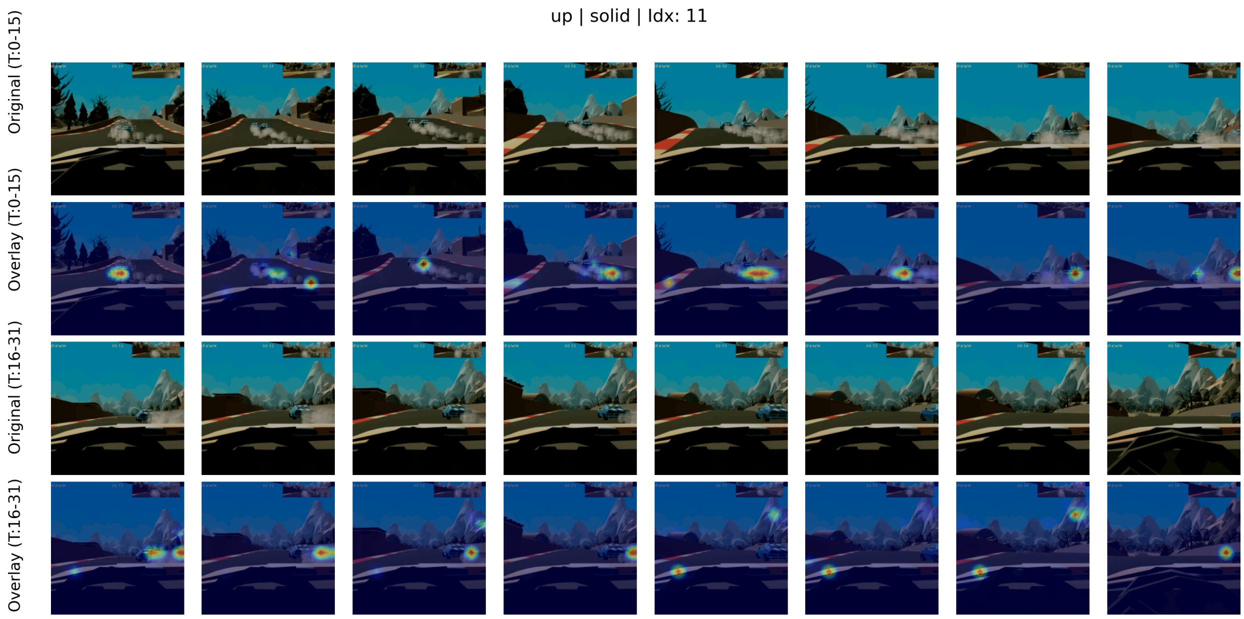


Figure 4: Interpretation example for an arousal-increasing segment in `solid`, generated by the model fine-tuned via **full fine-tuning** (first fold).

tion maps for other games and arousal classes, along with further comparative analyses, are available in our online repository.

References

[1] Y. Lee et al., "Surgical fine-tuning improves adaptation to distribution shifts," in *The Eleventh International Conference on Learning Representations*, 2023. [Online]. Available: `https://openreview.net/forum?id=APuPRxjHvZ`.